

Autonomous Agent-Driven Analysis of Federal Sentencing Disparities (FY2024)

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ABSTRACT

What explains persistent disparities in federal sentencing outcomes across race, citizenship status, and socioeconomic position—and can autonomous artificial intelligence systems meaningfully uncover these patterns within complex administrative data? Although the United States Sentencing Commission (USSC) provides detailed sentencing records, the technical complexity of these datasets limits systematic analysis. This study evaluates whether a large language model (LLM)-driven autonomous database agent can reduce analytic barriers and generate rigorous insights into contemporary federal punishment.

This research bridges the gap between traditional econometric sentencing studies, which require extensive manual preprocessing, and emerging AI-assisted legal analytics, which rarely demonstrate full deployment on real government datasets. A ReAct-style agent was developed and implemented to analyze the USSC Fiscal Year 2024 Individual Datafile, comprising over 60,000 cases. The system combines structured reasoning, read-only SQL execution, and schema interpretation assisted by retrieval to translate natural-language queries into reproducible analytical workflows.

The paper advances a dual claim. Methodologically, LLM-integrated agents can scale high-dimensional legal data analysis while maintaining transparency and control. Substantively, the FY2024 federal docket exhibits a bifurcated structure of punishment organized by citizenship and socioeconomic status, in which liberty and capital serve as substitutable forms of sanction. After reducing hundreds of administrative fields to 31 analytically relevant variables—including sentence type, incarceration status, race, citizenship, education, and fines—the agent tested three hypotheses: citizenship predicts incarceration; racial groups differ in exposure to custodial-only sentences; and education correlates with financial penalties. The findings reveal pronounced structural asymmetries and demonstrate the feasibility of AI-augmented legal analytics as a tool for surfacing systemic patterns in federal punishment.

Index Terms: *autonomous agents, federal sentencing, racial disparities, large language models, legal analytics, criminal justice, retrieval-augmented generation, socioeconomic stratification*

I. INTRODUCTION

The administration of federal justice in the United States generates a vast and intricate digital record. The USSC pub-

lishes annual Individual Datafiles that detail the characteristics of every sentenced federal felony and Class A misdemeanor, providing a comprehensive account of who the federal system punishes and how. The problem is that these records are difficult to use. Hundreds of variables carry obscure numerical codes whose meanings are buried in a technical codebook, and extracting a coherent picture of sentencing patterns requires substantial preprocessing before any substantive question can be asked.

This project introduces an alternative: an Autonomous Database Agent that does not merely query data but reasons about it. The system identifies missing values, formulates hypotheses, and executes Python and SQL code to test those hypotheses against raw records. The primary dataset is the FY2024 Individual Datafile. The analysis focuses on three dimensions of the federal sentencing system—the racial and citizenship composition of the docket, how sentence type varies across demographic groups, and how socioeconomic indicators relate to the imposition of financial penalties. The report serves a dual purpose: it is a methodological proof of concept for autonomous AI applied to legal analytics, and a substantive inquiry into the structure of federal punishment in the United States.

II. HYPOTHESIS FORMULATION

Before deployment, three working hypotheses were formulated to guide the automated inquiry, drawn from the prompt engineering strategy documented in the agent’s log files.

The first hypothesis (Hypothesis A) examines citizenship status as a determinant of sentencing outcomes. Non-U.S. citizens—particularly those classified as Illegal Aliens—are expected to experience higher incarceration rates (INOUT = 0) than U.S. citizens, on the reasoning that citizens possess stronger community ties and are therefore more eligible for probation or alternative sentencing.

The second hypothesis (Hypothesis B) addresses racial disproportionality in sentence type. The NEWRACE variable is expected to reveal statistically significant deviations in Sentence Imposed (SENTIMP), with minority groups—particularly Hispanic and Black defendants—disproportionately represented in the Prison Only category relative to White counterparts.

The third hypothesis (Hypothesis C) examines the relationship between education and financial penalties. The fine amount (AMTFINEC) is expected to correlate positively with

education level (NEWEDUC), on the logic that courts use educational attainment as a proxy for earning potential, producing heightened financial penalties for those perceived as higher-capacity or white-collar offenders.

III. DATA ENGINEERING AND FILTERING PIPELINE

A. Variable Selection Strategy

The raw USSC dataset contains hundreds of columns, many of which are administrative flags with no relevance to sociological analysis. A cleaning pipeline was implemented to reduce the dataset to a manageable form. The filtering script enforced strict schema reduction, retaining 31 high-value variables from the full field list. Target variables included SENTIMP (Sentence Type), INOUT (Prison vs. Non-Prison), AMTFINEC (Fine Amount), and MNTHDEPT (Months of Departure). Predictor variables comprised NEWRACE (Race), CITIZEN (Citizenship Status), NEWEDUC (Education), and AGE. A third set of quality-control variables—ACCCAT (Criminal History), DISPOSIT (Disposition), and DSPLEA (Plea Agreement)—was retained to assess data reliability and potential confounding.

B. Iterative Chunk Processing

Loading the entire dataset into memory was not feasible. Processing was constrained both by local RAM limits¹ and by token-rate limits imposed by Microsoft Azure for the billing tier assigned to the project. The filtering script addressed this by processing the file in chunks of 20,000 rows using the pandas library. This approach prevented RAM overflow by streaming data rather than loading it whole; it detected required columns dynamically in each chunk rather than assuming a fixed schema; and it forced all fields to string type on initial read, preventing type-inference errors and preserving the integrity of coded categorical variables. The resulting output—`filtered_cases_redone.csv`—served as the clean knowledge base for the agent.

IV. AUTONOMOUS AGENT ARCHITECTURE

A. The ReAct Logic Loop

The analytical engine was built using the LangChain framework and the Azure OpenAI API [2], implementing a ReAct (Reasoning and Acting) paradigm adapted from a DeepLearning.AI course on building database agents [3]. The loop operates in five stages. The LLM first receives a natural-language query—for instance, “analyze the relationship between race and prison sentences”—and reasons through its meaning and scope. It then decomposes the query into a sequence of logical steps: mapping variable names, formulating the necessary queries, determining execution order, and anticipating how results should be interpreted. Having formed that plan, the agent invokes specific tools to carry out each step. It observes the output—a pandas DataFrame or an SQL result set—and extracts pertinent information. Finally, it synthesizes the

gathered data into a coherent textual response. This cycle of thought, planning, action, observation, and synthesis maintains a clear audit trail of how each conclusion was reached, keeping the agent’s reasoning legible rather than opaque.

B. SQLite Integration

To enable complex aggregation, the filtered CSV was ingested into a local SQLite database. This design is preferable to pure pandas manipulation for two reasons. SQL is optimized for GROUP BY and JOIN operations on disk-based storage, substantially reducing the agent’s memory footprint during analytical queries. All database access was also governed by a strict read-only policy: the execution function permitted only SELECT, PRAGMA, EXPLAIN, and WITH statements, preventing any accidental modification of the underlying data.

C. Retrieval-Augmented Generation Codebook

A critical design challenge was variable opacity. USSC codes are not self-explanatory—NEWRACE = 6 carries no inherent meaning to a language model. To address this, a retrieval-augmented generation (RAG) module was implemented. The USSC Public Release Codebook (FY1999–FY2024) was ingested and vectorized using a TF-IDF vectorizer with an n-gram range of (1, 2). When the agent encountered an ambiguous column, it triggered a lookup function that computed cosine similarity between the query term and vectorized codebook chunks, returning the precise definition.

In practice, the approach encountered significant limitations. PDF parsing failed almost entirely, leaving the agent unable to map natural-language terms such as “race” or “age” to their corresponding column identifiers. A query as straightforward as “what is the distribution of race across age” would fail at the schema-resolution step. The consequence was that every prompt required a manually prepended explanation of the relevant column codes, drawn directly from the codebook. The RAG module thus functioned as a conceptual scaffold rather than a working automation. This failure is itself informative: it marks a technical ceiling that future implementations must address before agent-based legal analytics can operate without expert supervision.

V. RESULTS

A. Data Integrity and Missingness

The agent’s initial quality sweep revealed critical gaps in variable availability. ACCCAT (Criminal History Category) and RANGEPT were missing in 59,550 records, rendering them statistically unusable and eliminating the possibility of controlling for prior criminal history in this analysis. The INOUT variable—central to determining incarceration status—was absent in 47,875 cases, suggesting it is recorded conditionally, perhaps only for guideline-sentence cases or specific districts. Where present, it provided a strong signal. By contrast, NEWRACE was missing only 446 values and CITIZEN only 336, producing completion rates above 99% and validating the demographic conclusions that follow.

¹VSCode has a RAM ceiling of 2,048 MB shared across the application and its extensions, making single-pass processing of approximately 1.6 GB impractical.

TABLE I
RACIAL DISTRIBUTION OF FEDERAL DEFENDANTS, FY2024.

Racial Category	Defendants	Share (%)
Hispanic (NEWRACE = 3)	30,472	49.8
Black (NEWRACE = 2)	14,997	24.5
White (NEWRACE = 1)	12,866	21.0
Other (NEWRACE = 6)	2,897	4.7

B. Demographic Distribution of Federal Crime

The agent's analysis of NEWRACE exposed a pronounced skew in the federal docket. Hispanic defendants (NEWRACE = 3) accounted for 30,472 cases, Black defendants (NEWRACE = 2) for 14,997, White defendants (NEWRACE = 1) for 12,866, and all other categories (NEWRACE = 6) for 2,897. The full distribution is presented in Table I.

Hispanic defendants account for approximately half of all processed federal cases. White defendants, who constitute the majority of the overall U.S. population, represent only 21% of the docket. This disparity is not readily explained by population-level criminality. It suggests instead that federal enforcement priorities in FY2024 were heavily concentrated on offenses with a disproportionate impact on Hispanic populations—most plausibly, immigration-related crimes under Title 8.

C. Sentencing Disparities: The Prison-Only Norm

Cross-tabulation of SENTIMP by NEWRACE provided granular insight into judicial severity across demographic groups. The Other racial category (NEWRACE = 6) recorded the highest prison-only rate in the dataset: 85.95% of individuals in this group received a Prison Only sentence. This figure likely reflects the jurisdictional circumstances of groups such as Native Americans prosecuted under federal rather than tribal law, or specific foreign nationals subject to mandatory sentencing structures with no alternative disposition available.

Among Hispanic defendants, the volume of cases combined with the documented prevalence of Prison Only dispositions confirms that the federal system functions, in practical terms, as a mechanism for mass custodial processing of this demographic. The agent noted that the majority of individuals sentenced to prison only were drawn predominantly from the Hispanic category, consistent with the immigration-enforcement composition of the docket described above.

D. The Citizenship-Incarceration Nexus

The investigation into CITIZEN and INOUT produced the most polarized results of the study, summarized in Table II. U.S. citizens were incarcerated in approximately 53% of cases where INOUT was recorded, with the remaining 47% receiving non-custodial outcomes such as probation, home confinement, or fines—something approaching an even probability of avoiding prison. Illegal Aliens were incarcerated in only 6.74% of recorded cases, with 93.26% receiving a non-custodial outcome, a pattern consistent with this population

TABLE II
INCARCERATION RATES BY CITIZENSHIP STATUS, FY2024.

Citizenship Category	Incarcerated (%)	Not Incarcerated (%)
U.S. Citizen (CITIZEN = 1)	53.04	46.96
Illegal Alien (CITIZEN = 3)	6.74	93.26
Extradited Alien (CITIZEN = 5)	0.37	99.63

being processed for deportation rather than domestic incarceration. Extradited Aliens produced the most extreme result: only 0.37% received a custodial outcome under the PRISDUM variable,² with 99.63% classified as non-custodial—consistent with the procedural reality that extradited individuals are typically returned to their home country or credited for time already served abroad while contesting extradition.

E. The Socioeconomic Cost of Justice

The agent's calculation of average AMTFINEC by NEWEDUC revealed a steep gradient across educational levels. Defendants with less than a high school education paid an average fine of \$161.79. College graduates paid an average of \$53,733.11—a difference of approximately 332 times. Federal sentencing guidelines for fines are explicitly calibrated to a defendant's ability to pay, and educational attainment functions in practice as a proxy for that capacity. Uneducated defendants receive nominal financial penalties because they are effectively insolvent. Educated defendants—more likely to be implicated in white-collar offenses such as fraud or regulatory violations, where the monetary gain and thus the sanctionable amount are substantial—face penalties commensurate with that gain. The result is a system in which the financial instrument of punishment scales sharply with economic standing.

VI. SPECULATIVE INTERPRETATION

The following section extrapolates potential causal mechanisms behind the statistical correlations identified above. These interpretations apply criminological theory to the agent's findings and are explicitly speculative.

A. The Immigration Engine Hypothesis

The USSC docket appears to conflate two distinct categories of federal action: traditional criminal enforcement addressing theft, violence, and drug offenses, and administrative immigration enforcement under Title 8. The 50% representation of Hispanic defendants and the high Prison Only rate for that group are likely not reflections of elevated violent recidivism but of mandatory minimums for illegal re-entry. Such sentences are often short—frequently time already served in detention—but they register statistically as imprisonment, inflating custodial metrics for this demographic while obscuring the administrative rather than correctional nature of the confinement.

²PRISDUM was substituted for INOUT in the Extradited Alien category as it was more suitable for capturing custody status within that subgroup.

B. The Extradition Anomaly

The 99.6% non-custodial rate for Extradited Aliens requires procedural explanation. The non-custodial code most plausibly captures individuals who are not placed in Bureau of Prisons custody—a category that applies to defendants transferred to U.S. Marshals for immediate deportation upon conviction, or those whose U.S. sentence is shorter than the period already spent in foreign detention while fighting extradition. In such cases, the defendant is released immediately upon judgment, producing a disposition that registers as non-custodial without representing meaningful freedom from punishment.

C. The Bifurcated Justice System

The fine and incarceration data together reveal two distinct systems of punishment operating within a single institutional framework. The first processes defendants of low educational attainment—predominantly minority and non-citizen—and extracts punishment in the form of time. Incarceration rates are high; fines average \$161 because the defendants are effectively insolvent. The second processes defendants of higher educational attainment—predominantly citizens—and extracts punishment in the form of money. Fines exceed \$53,000 on average, while incarceration rates are comparatively lower. Education in this dataset functions as a proxy for defensibility: educated defendants can retain private counsel who negotiate plea agreements involving fines and probation, while less educated defendants rely on public defenders and face the rigid Prison Only sentencing grid. Liberty and capital are, in this reading, substitutable currencies of punishment whose allocation is determined largely by socioeconomic class.

VII. LIMITATIONS

Several significant constraints limit what can be concluded from this analysis. The ACCCAT variable, which encodes criminal history complexity, was missing across nearly the entire dataset. Without it, the analysis cannot control for career-offender status—a primary driver of sentencing severity—leaving open the possibility that demographic disparities in incarceration reflect prior criminal history rather than identity-based inequity.

The high missingness in INOUT (over 47,000 rows) introduces a risk of selection bias. If INOUT is recorded only for guideline cases and not for plea agreements, or vice versa, the incarceration rates reported above could be systematically skewed. The agent’s interpretation of INOUT codes also relies on RAG retrieval of the codebook; an ambiguous definition could invert the directional conclusions about citizenship and incarceration, though the agent’s explicit output—stating that 0 equals receipt of a prison sentence—was accepted as ground truth.

The fine analysis is subject to ecological fallacy. The finding that educated defendants pay higher fines describes aggregate group behavior, not individual outcomes, and the group means are substantially driven by high-value outliers—executives fined millions following fraud convictions—that pull the average far above what a typical educated defendant would face. Finally, the restriction to 31 columns may have excluded

variables with greater explanatory relevance to the hypotheses tested; a more exhaustive variable selection strategy would strengthen the robustness of future iterations.

VIII. CONCLUSION

Three findings define the FY2024 federal sentencing landscape as revealed by the agent. The federal docket is predominantly processing Hispanic individuals—approximately half of all cases—a composition that reflects administrative immigration enforcement as much as traditional criminal prosecution. The system exhibits a stark substitution between liberty and capital as instruments of punishment: defendants with less human capital pay with their freedom, those with more pay with their finances. And citizenship is the single most potent predictor of whether a defendant receives a custodial outcome, with non-citizens overwhelmingly processed through pathways that terminate in deportation rather than domestic incarceration.

The methodological findings are equally clear. An LLM-driven agent can navigate a complex, partially obfuscated government schema and surface policy-relevant patterns at a fraction of the time required by manual econometric preprocessing. The RAG codebook failure—where PDF parsing collapsed and the agent could not resolve variable names without manual intervention—marks the ceiling of current capability and the next problem to solve. Until schema interpretation is reliable without expert input at the prompt stage, agent-based legal analytics will require supervision. With reliable schema resolution, this methodology could become a genuinely accessible tool for researchers and policymakers who need to interrogate large administrative datasets without the overhead of traditional quantitative methods.

REFERENCES

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APPENDIX

Due to a recording error during execution, the SQL queries and raw agent output were not fully captured. A model demonstration is available on request. Representative raw queries and output are archived at: https://docs.google.com/document/d/1K4IM9vSMO5SnDIutEwlje3UbUbs88PNRVF8svzuDto/edit?usp=drive_link